Urban Climate Dashboard

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1. Problem Statement

Urban areas worldwide face rapid climate change due to industrialization and urbanization. Raw monthly climate data across 20 cities and 45 years (1980–2025) is difficult to interpret without visualization. This project builds an interactive web dashboard to:

- Track long-term climate trends for individual cities (temperature, humidity, precipitation, wind, UHI)
- Compare climate metrics across cities and continents for any selected year
- Reveal seasonal patterns through monthly breakdowns
- Present continent-level distribution using a donut chart

2. Dataset Overview

Field	Description	Unit
temperature_celsius	Average monthly temperature	°C
humidity_percent	Average monthly humidity	%
precipitation_mm	Monthly precipitation	mm
wind_speed_ms	Monthly wind speed	m/s
urban_heat_island_intensity	UHI vs rural areas	Index
city / continent	20 cities across 4 continents	Text
year / month	Time period covered	1980–2025

3. Dashboard Controls

3.1 Interactive Controls

Control	Function
City Selector (dropdown)	Pick any of 20 cities — all charts update instantly
Metric Selector (dropdown)	Switch between Temperature, Humidity, Precipitation, Wind, or UHI
Continent Filter (dropdown)	Narrow charts to Americas, Asia, Europe, or Africa
Year Slider (1980–2025)	Scrub any year — stat cards, bar, donut & scatter update live
5 Stat Cards (top of page)	Show city's exact values for all 5 metrics for selected year

3.2 Stat Cards

Five summary cards appear at the top of the dashboard, showing the selected city's average values for the selected year across all five metrics. Each card is color-coded and updates dynamically when the city or year changes.

4. Dashboard Graphs & Insights

Graph 1 — Annual Trend Line Chart

Shows how a climate metric changes year by year (1980–2025) for the selected city. A gradient fill under the line highlights the trend area clearly.

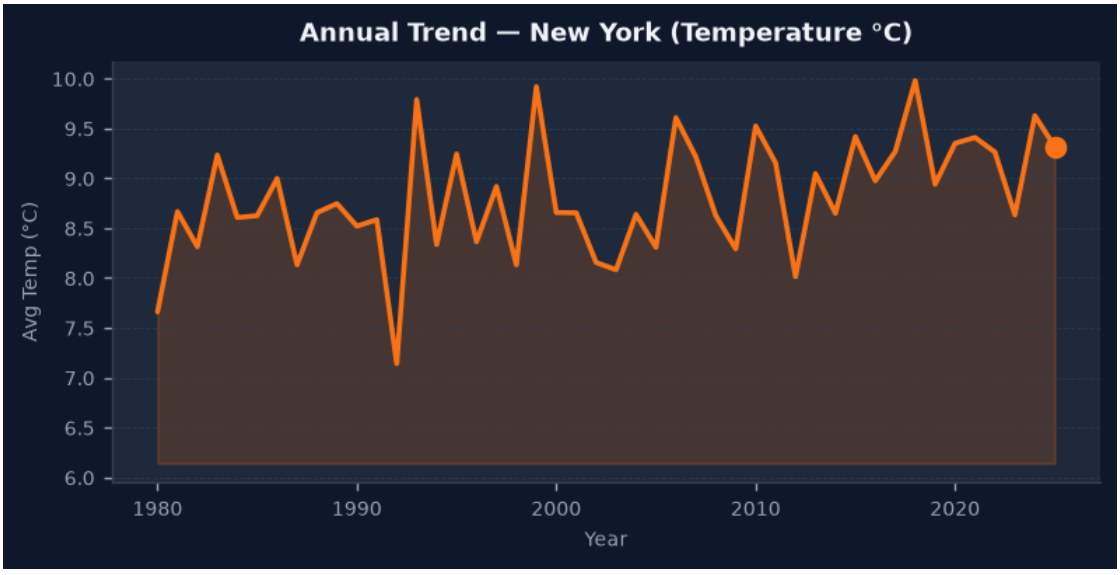


Fig 1: Annual temperature trend for New York (1980–2025)

Key Insight:

Global average temperature rose from 12.98°C (1980) to 13.90°C (2025) — a +0.93°C increase over 45 years, confirming the effect of climate change across all cities.

Graph 2 — Monthly Profile Chart

Displays month-by-month variation (Jan–Dec) of the selected metric for the chosen city and year. Temperature is shown as a line; precipitation as bars for easy comparison.

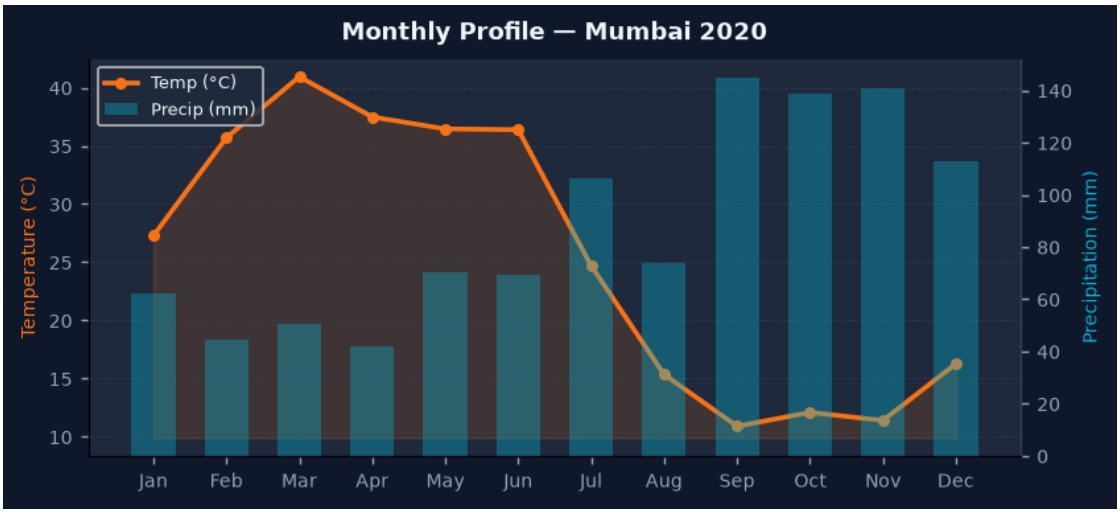


Fig 2: Monthly temperature & precipitation for Mumbai (2020)

Key Insight:

Mumbai and Delhi show sharp precipitation spikes in June–August (monsoon season) while January–February are nearly dry. April is the warmest month globally (28.47°C avg); October is the coldest (-1.53°C avg).

Graph 3 — City Comparison Bar Chart

Ranks all cities by the selected metric for the chosen year. Each city has a unique color and gradient bar. Can be filtered by continent using the dropdown.

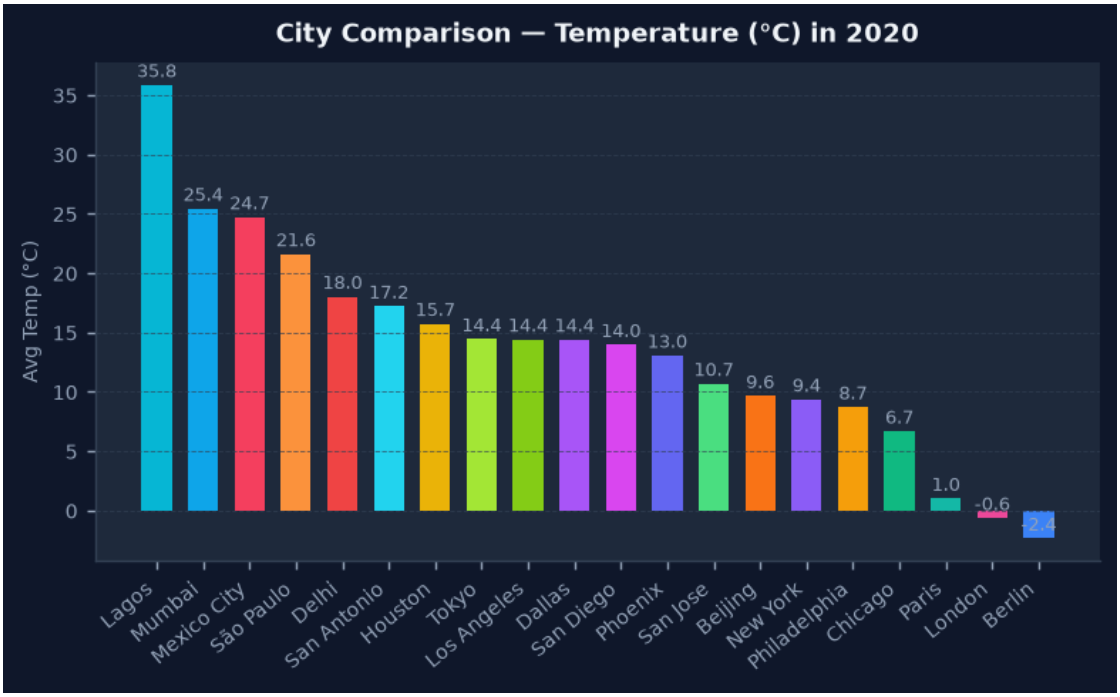


Fig 3: City comparison — Temperature (°C) for year 2020

Key Insight:

Lagos dominates at 35.15°C — nearly 10°C hotter than 2nd-place Mumbai (25.29°C). Berlin (-1.59°C), London (-0.83°C), and Paris (1.26°C) are the coldest. For precipitation, Lagos (81.4 mm) and Mumbai (81.0 mm) lead — nearly double European cities.

Graph 4 — Continent Share Donut Chart

Shows the proportional average of the selected metric split across 4 continents for the chosen year. Slice size = continent's average value. Centre shows the combined average.

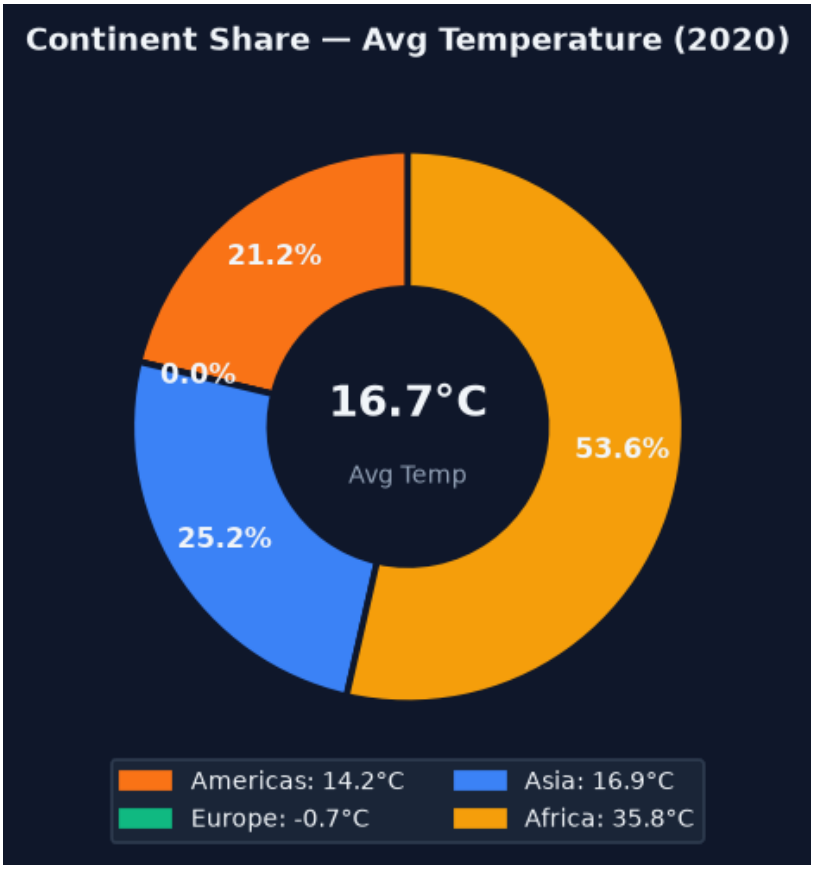


Fig 4: Continent share of average temperature (2020)

Key Insight:

Africa's slice dominates at 35.15°C while Europe's slice is nearly invisible at -0.39°C — a gap of over 35°C between the two continents. This visually highlights the massive thermal divide between regions.

Graph 5 — Temperature vs Humidity Scatter Plot

Places each city as a dot in 2D space — X axis = avg temperature, Y axis = avg humidity. Each city has a unique color. Reveals clusters and correlations between the two metrics.

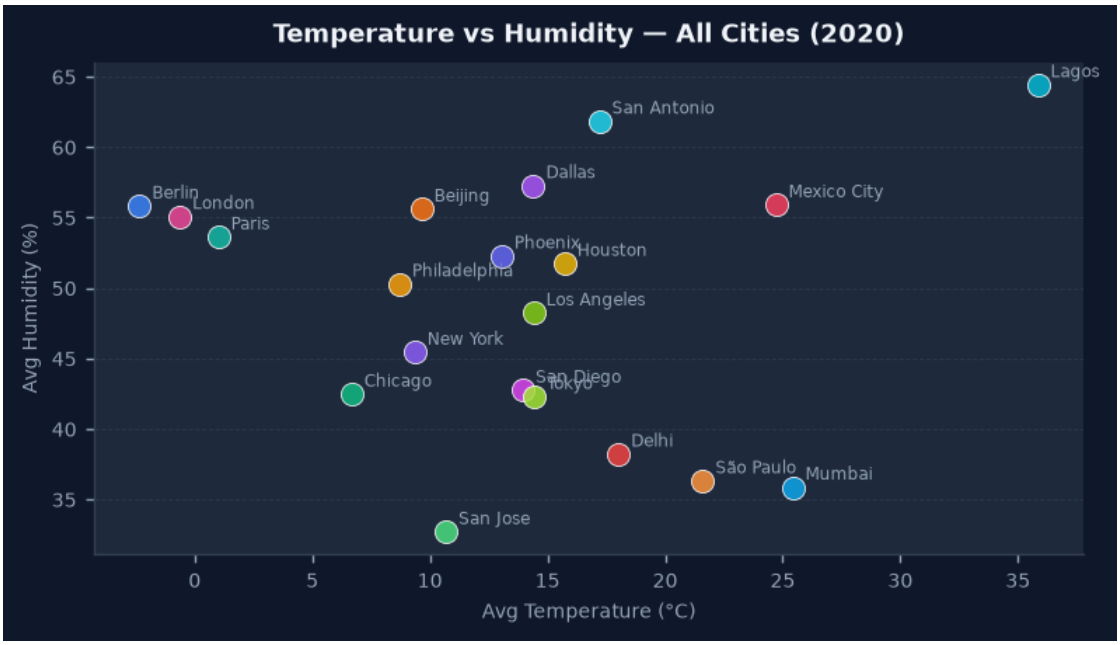


Fig 5: Temperature vs Humidity scatter for all cities (2020)

Key Insight:

Lagos appears top-right — hottest AND most humid (62.9%). Phoenix and Delhi are hot but relatively dry. European cities cluster bottom-left (cold, moderate humidity). No strict linear correlation exists between temperature and humidity globally.

5. Key Insights Summary

Insight 1: Global Temperature Rise (Climate Change Evidence)

Insight 2: Lagos is the Hottest City — by a Large Margin

Insight 3: European Cities are the Coldest

Insight 4: Urban Heat Island Effect is Significant

Insight 5: Monsoon Cities Receive the Highest Rainfall

Insight 6: Lagos is Also the Most Humid City

Insight 7: Seasonal Patterns Vary Dramatically by City

Insight 8: Africa vs Europe — A 35°C Temperature Gap

Insight	Finding	Source Chart
Global Warming	+0.93°C rise from 1980 to 2025	Annual Trend Line Chart
Hottest City	Lagos — 35.15°C average	Bar Chart + Donut Chart
Coldest City	Berlin — -1.59°C average	Bar Chart + Donut Chart
Most Humid	Lagos — 62.9% humidity	Bar Chart + Scatter Plot
Most Rainfall	Lagos — 81.4 mm/month	Bar Chart + Monthly Chart
Hottest Month	April — 28.47°C global average	Monthly Profile Chart
Coldest Month	October — -1.53°C global average	Monthly Profile Chart
Urban Heat Island	2.0 intensity in NY, Tokyo, Delhi etc.	Stat Cards + Trend Chart
Continent Gap	Africa 35.15°C vs Europe -0.39°C	Donut Chart + Bar Chart
Monsoon Pattern	Mumbai/Delhi peak precip in Jun–Aug	Monthly Profile Chart

6. Tools & Technologies used

Tool	Purpose
Python (csv, json, zipfile)	Data loading, processing, aggregation & export
HTML5 + CSS3	Dashboard page structure and styling
HTML5 Canvas API	Drawing all 5 charts from scratch — no library used

7. Conclusion

The Urban Climate Dashboard converts 11,040 rows of raw climate data into a fully interactive visual experience — exposing climate change evidence, continent-level disparities, monsoon patterns, and the Urban Heat Island effect. Built entirely from scratch using Python and JavaScript Canvas API, the final output is a single portable HTML file that works offline in any browser.